

EMBRYONIC DEVELOPMENT

5. FIRST WEEK OF EMBRYONIC DEVELOPMENT

Human embryogenesis (or embryogeny) refers to the process of development of the human embryo from fertilization until the fourth week of development. The period from the fourth to the tenth week of development is referred to as organogenesis.

We speak of an embryo from fertilization to the eighth week of embryonic development, after which we will speak of a fetus until birth.

In humans, the first week of development corresponds to the 3rd week of the mother's menstrual cycle. It includes two fundamental phenomena: fertilization and segmentation.

1. FERTILIZATION

Fertilization is the initial stage in the formation of the individual. It is the formation of the egg or zygote by the fusion of the two gametes, paternal and maternal. It defines time 0 of embryonic development.

In humans, fertilization takes place in the outer third of the fallopian tube. It is preceded by the movement of the gametes, their conditioning, and their union. These essential stages occur within the female reproductive tract.

1.1. GAMETE CONNECTING

Sperm path

During each sexual encounter, at the time of ejaculation, 100 to 300 Millions of sperms are deposited in the vaginal cavity. Among these , 40 to 70%, are mobile. They have to go back up almost the entire female reproductive tract to reach the ampulla of the fallopian tube, where the female gamete is located. From there, they penetrate the membranes surrounding the oocyte. The entire process is rapid: the first sperm reach the ampulla of the fallopian tube within 30 minutes. The environment modifies the sperm and prepares them for fertilization. During this migratory stage, two phenomena occur:

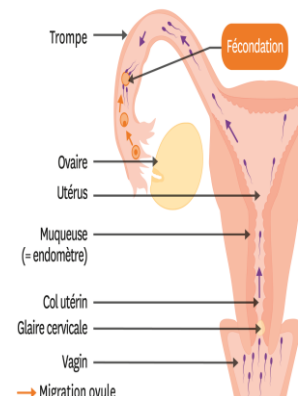


Figure 1. Path of sperm

a. Reduction in sperm count

There is a considerable loss of sperm: only a few hundred will reach the oocyte. In addition to the 25% of abnormal sperm, most, failing to complete their migration, are destroyed on site by phagocytic cells in the female reproductive tract.

b. Capacitation

This refers to all the membrane modifications that make sperm capable of fertilization. Capacitation involves the removal of the cell coat (glycoproteins) deposited during epididymal transit and a redistribution of membrane proteins. This results in an increase in sperm metabolism.

1.2. CROSSING THE OVOCYTE ENVELOPES

The sperm must overcome several obstacles before reaching the cytoplasm of the oocyte.

a. Passage of the Cumulus Oophorus and Corona Radiata

In the cumulus nucleus, the cells are more or less dissociated, and passage is still possible. Sperm cells infiltrate the intercellular spaces. It is the outermost cell layer of the corona radiata, surrounding the zona pellucida, that requires enzymatic dissociation. This passage, which lasts a few minutes, is facilitated by an enzyme, acrosome hyaluronidase.

b. Passage of the pellucid membrane

The passage is species-specific. Sperm recognition within the species occurs thanks to the presence, in the zona pellucida, of proteins possessing receptor sites for the sperm plasma membrane. The zona pellucida attracts and retains the sperm and induces the acrosomal reaction. Sperm capture is facilitated by a zona pellucida protein: the ligand ZP3, an receptors located in the membrane of Sperm (Gal Tase). Enzyme release originating from the acrosome, acrosin, allows the penetration of the membrane by the sperm.

c. Acrosomal reaction

This is the opening of the acrosome. It cannot happen that after capacitation and requires the presence of Ca^{++}

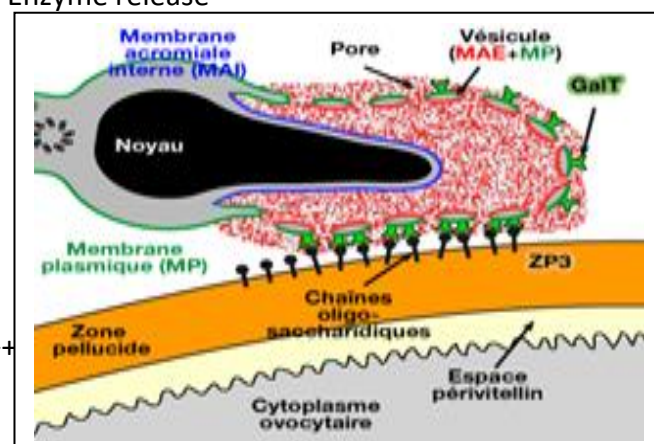


Figure 2. passage throuthe
pellucida membarne

The acrosomal reaction, which is a phenomenon of exocytosis, is induced by the protein fraction of ZP3 in the zona pellucida. It consists of the progressive fusion of the plasma membrane

and the outer membrane of the acrosome, until giving rise to Membranous vesicles. This fusion creates pores allowing the release of the acrosome's contents. This reaction is rapid, It only lasts 10 to 15 minutes and it is the inner membrane of the remaining acrosome which limits the anterior two-thirds of the head of the sperm, the plasma membrane remains intact at the back.

1.3. STAGES OF FERTILIZATION

Fertilization begins when the sperm comes into contact with the plasma membrane of the oocyte, in the peri-oocyte space. Several events then occur:

a. Membrane fusion phenomenon

For the fusion of gametes (fusion of cytoplasmic membranes = plasmogamy) In the human species, the sperm lies flat on the surface of the oocyte.

The two membranes adhere and then fuse. plasma (fusion takes place in the region post-acrosomal, where the membrane persists (plasma of the sperm). Adhesion occurs membrane adhesion proteins are involved such as integrin and fertiline.

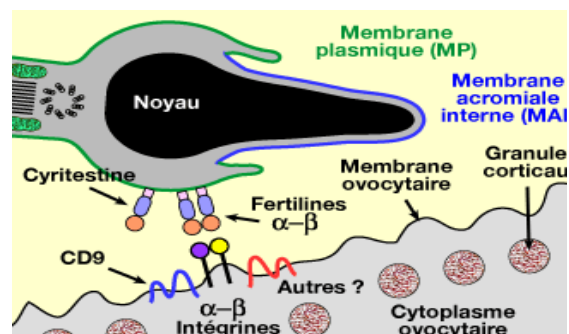


Figure 4. Membrane fusion phenomenon

The still-motile sperm penetrates the oocyte's cytoplasm and eventually becomes fully incorporated into the oocyte, including its flagellum but without its plasma membrane, which remains outside. This membrane fusion immediately causes membrane depolarization, which propagates within seconds. It is due to the influx of Na^+ through a change in membrane permeability. This prevents the adhesion of another sperm. It is one of the mechanisms that prevents polyspermy.

b. Activation of the egg

An increase in intracytoplasmic Ca^{++} and pH triggers egg activation, which manifests itself through three phenomena:

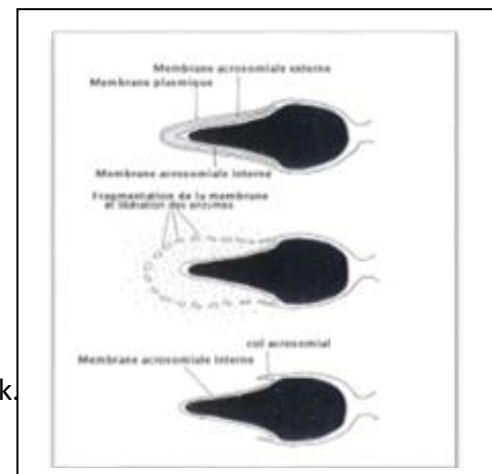


Figure 3. The acrosomal reaction

- **Cortical reaction**

A rapid cortical reaction occurs within 15 to 20 minutes, with the formation of numerous vesicles; the cortical granules open at the surface, releasing substances beneath the affected area.

pellucida of the periovular fluid, forming around the oocyte a space separating the oocyte from the zona pellucida.

The contents of the granules (enzymes: hydrolases and peroxidases) appears to alter the composition of the zona pellucida and could explain the blockage of the sperm present at the periphery. The modified zona pellucida (membrane of fertilization) would thus ensure the control of monospermia.

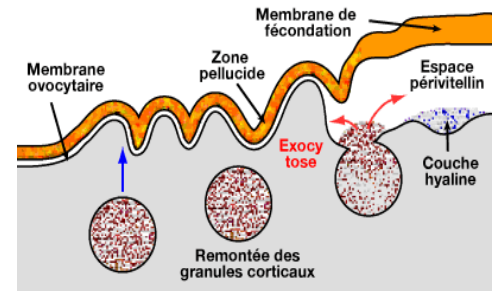


Figure 5. The cortical reaction

- **Nuclear reaction**

It is also rapid and occurs within 20 minutes. This is the end of meiosis in the female gamete (the second meiotic division), which is only completed after fertilization. The secondary oocyte becomes an ovum.

- **Cytoplasmic reaction**

Activation of the cytoplasm accompanies the cortical reaction: metabolism increases, with a particularly significant synthesis of RNA. Protein synthesis resumes and cellular respiration increases. These phenomena result in a large cell composed of the cytoplasm of the secondary oocyte, now an ovum, containing two nuclear elements: one originating from the oocyte (female pronucleus) and the other from the sperm (male pronucleus). Simultaneously, the sperm's cytoplasm becomes disorganized. It is generally accepted that it is completely reabsorbed and plays no role in the activation of the egg.

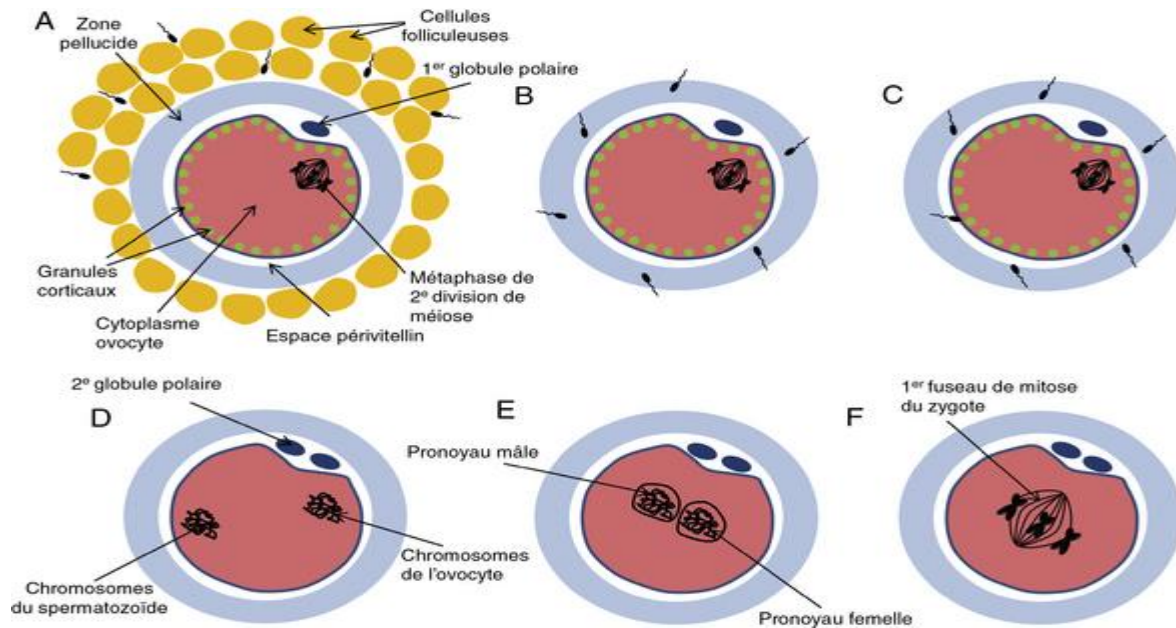


Plate 1. Stages of fertilization

c. Amphimixia or caryogamy

This is the fusion of the two haploid sets of chromosomes. It only occurs after the formation of the two pronuclei. It will be immediately followed by the first cell division of the egg, resulting in the formation of the first two blastomeres.

The two male and female pronuclei appear simultaneously, approximately 10 hours after the fusion of the gametes. Near the nucleus of the sperm, its proximal centriole. It induces the formation of an achromatic apparatus, the sperm aster which pulls the male pronucleus towards the center of the ovum and the female pronucleus also approaches the center of the ovum.

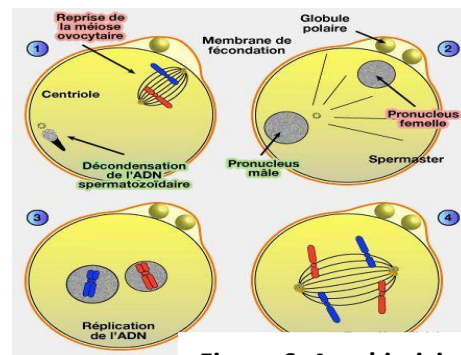


Figure 6. Amphimixia

At the center of the ovum, the two pronuclei adhere closely. The chromosomes gradually condense. The pronuclei's outer membrane is resorbed, and the condensed chromosomes align in a plane within an achromatic spindle. This marks the transition to metaphase of the first mitosis, the cleavage of the egg, which occurs approximately 20 to 24 hours later, culminating in gametic fusion that produces the first two blastomeres.

1.4. CONSEQUENCES OF FERTILIZATION

- **Reconstitution of a diploid number of chromosomes**

Half of the cells originate from the male pronucleus, the other half from the female pronucleus. The nucleus of each blastomere contains 44 autosomes and 2 sex chromosomes.

- **Determining the sex of the zygote**

Which results from the sex chromosome contained in the sperm involved in fertilization.

- **Creation of a new individual**

Through the mixing of parental genomes. There is transmission of hereditary traits, but also gene shuffling during meiosis. This is the purpose of sexual reproduction.

2. SEGMENTATION

The cleavage process affects the entire egg. It begins immediately after fertilization. It is an uninterrupted series of mitoses that break the egg into smaller daughter cells, the blastomeres.

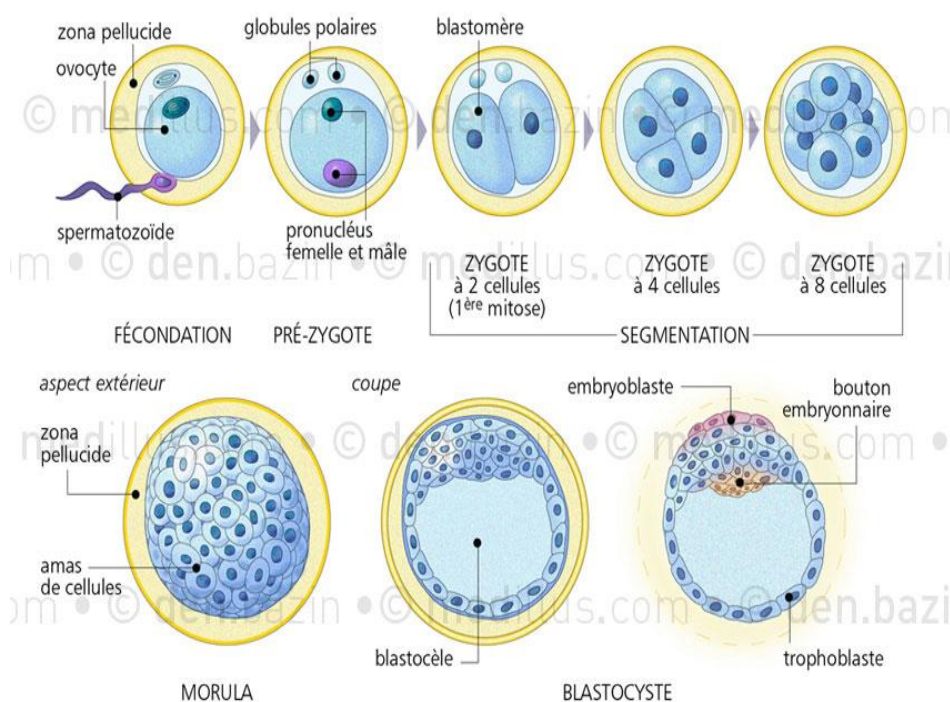


Figure 7. Segmentation

2.1. CHARACTERISTICS OF SEGMENTATION

Segmentation in humans is said to be:

- Asynchronous: the divisions are not completely synchronous. Between the stages with 2, 4 or 8 blastomeres, there are odd-numbered stages with 3, 5, 6 and 7 blastomeres.
- Unequal: during each division, one of the blastomeres is slightly larger than the other. It will divide first.
- Total: The entire egg is segmented.

During segmentation, the following stages are observed:

- | | |
|---|---|
| - Stage 2 blastomeres between 24 and 30 hours | Blastomeres are ovoid, independent cells, |
| - Stage 3 blastomeres between 30 and 36 hours | separated by relatively large intercellular spaces; |
| - Stage 4 blastomeres between 36 and 40 hours | the first blastomeres are totipotent (are |
| - Stage 8 blastomeres between 40 and 50 hours | equivalent and have the same potentialities, and |
| | only one of them can give rise to an entire |
| | organism). |

2.2. MORULA STAGE

The term morula refers to the solid mass of cells formed after four or five successive divisions of the blastomeres. The morula stage (32 to 64 blastomeres) occurs between the third and fourth day. The egg is surrounded by the zona pellucida and its size does not increase. At the end of this stage, the outer cells are smaller and darker than the inner cells. They have flattened, and the surface of the morula is smooth.

2.3. COMPACTION PHENOMENON

Before the 32-blastomere stage, the blastomeres are spherical, all visible, and can be separated from one another. From this stage onward, the blastomeres flatten, increasing their surface area in contact with neighboring cells through gap junctions. This is compaction. It is the first differentiation of the cells. The tight junctions between the peripheral cells control the exchange of substances between the morula and the external environment.

2.4. BLASTOCYST STAGE

When the morula reaches approximately 60 cells or more, gaps appear between the inner cells, near one pole. These gaps fuse into a single cavity, the blastocoel, which contains fluid

filtered by the cells from the uterine cavity. The morula then becomes a blastocyst. Two groups of cells can be distinguished within it:

- **The trophoblast:** simple peripheral layer, formed of small flattened cells (micromeres). It will give rise to the placenta and the membranes.
- **The embryonic button (embryoblast):** The inner cell mass, composed of larger, polyhedral or rounded cells (macromeres), is the embryonic disc. It is attached on one side to the inner surface of the trophoblast and, on the other side, in contact with the blastocoel. It will give rise to the embryo proper and its extraembryonic structures.

The two groups of cells differ in their synthetic activity, their mitotic rhythm, their shape and also in their ultra-structural characteristics.

The cells of the embryoblast accumulate at one pole; this is the embryonic pole of the blastocyst as opposed to the opposite pole (anembryonic): this is the polarity of the embryo.

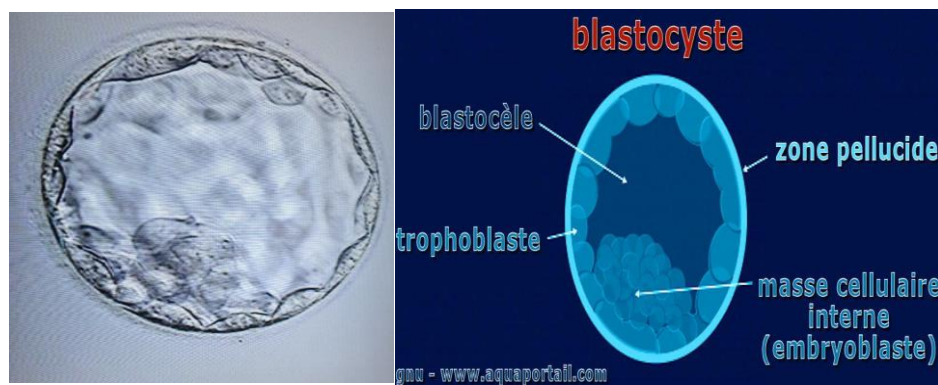


Figure 8. Structure of the blastocyst

2.5. BLASTOCYST ECLOSION (EXPENDED BLASTOCYST)

On the 5th day, the zona pellucida thins and weakens due to enzymatic action. This enzymatic action, combined with intracavitary pressure from sodium flow, causes the membrane to rupture, and the egg emerges, deformed.

From this point on, the blastocyst can grow by volume. Its diameter ends of the 6th day.

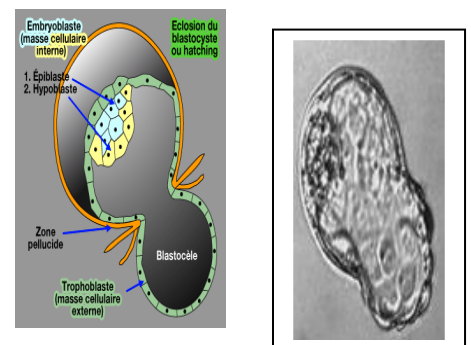


Figure 9. Expanded Blastocyst

2.6. Tubal Migration

During cleavage, the egg is passively transported from the ampulla of the fallopian tube to the uterine cavity. Depending on the egg's transit through the ampulla, several stages are observed:

- The 2 to 3 blastomere stages: the tubal isthmus.
- The 4 to 8 blastomere stages: the interstitial segment.
- The 12 to 16 blastomere stages: the uterine isthmus.

So it is a morula that arrives in the uterine cavity. The blastocyst lives freely and then makes contact with the endometrium on the 6th day.

3. Pre-implantation

- From day 6 to day 7, the blastocyst comes into contact with the endometrium through its embryonic pole.
- Upon adhesion, trophoblast cells proliferate at the point of attachment, forming a syncytial cell mass called the syncytiotrophoblast (ST), which originates from the fusion of outer trophoblast cells. The syncytiotrophoblast is a highly active, rapidly proliferating tissue that produces proteolytic enzymes and secretes factors that enable it to induce apoptosis of uterine mucosal epithelial cells, cross the basement membrane, and penetrate the underlying stroma, coming into contact with uterine blood vessels.
- The rest of the trophoblast is still made up of well-defined cells. It is the cytotrophoblast (CT) that covers the blastocoel and separates the embryonic disc from the syncytiotrophoblast.
- The trophoblast produces the hormone chorionic gonadotropin or hCG. This polypeptide hormone produced by the embryo causes the maintenance of the ovarian corpus luteum (luteotropic action) and ensures the interruption of cycles. The corpus luteum becomes a corpus luteum of pregnancy and continues to produce estrogen and progesterone for more than ten weeks; this function is then taken over by the placenta. The human embryo, through its hCG levels, controls its own implantation.

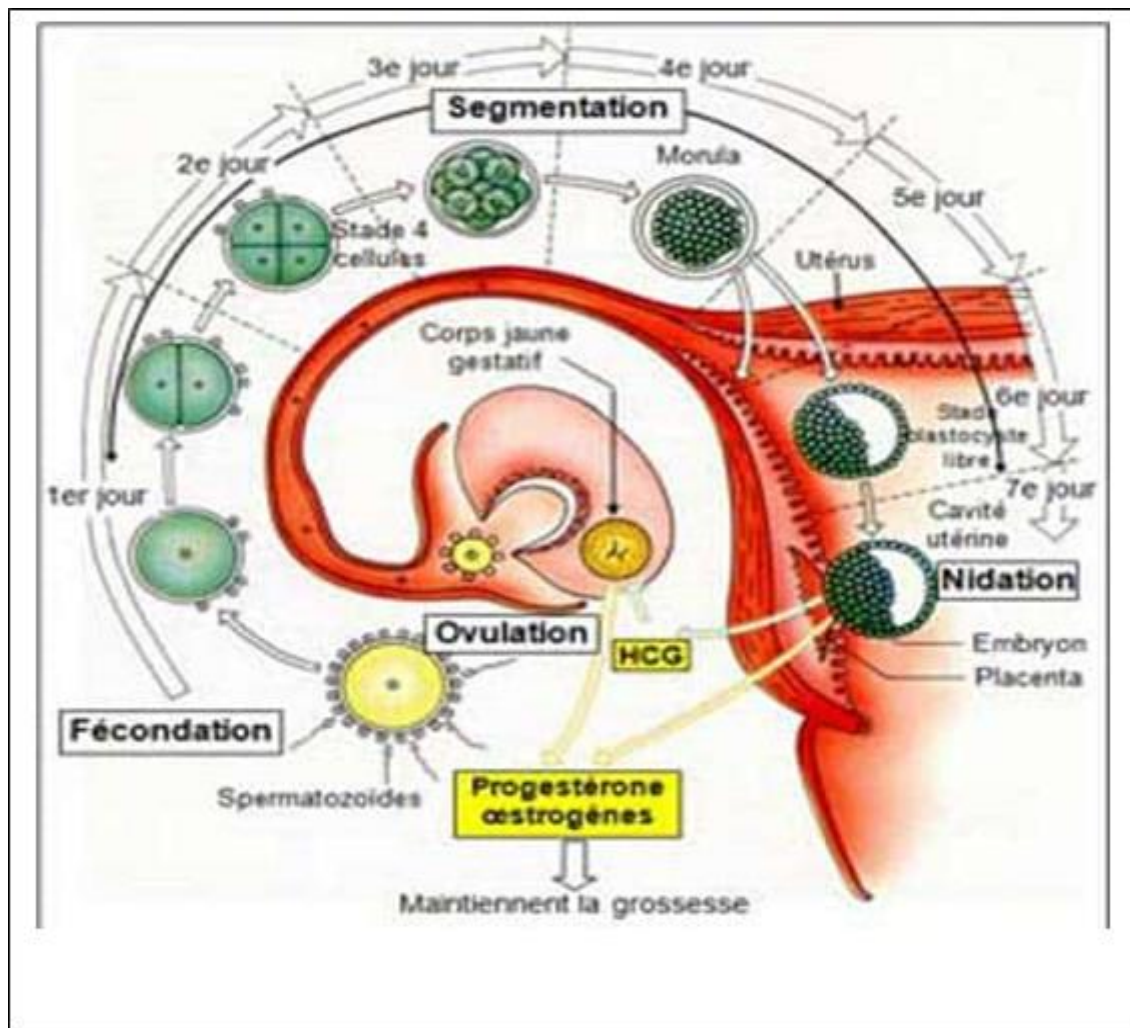


Figure 10. Cleavage and tubal migration